

Why Bias reduced with weighted average (Ada Boost)

- Ada Boost labels take values of -1 or 1
 - weight of a sample to be chosen changes
1. Initialize weights for samples $w_i^{(1)} = \frac{1}{N}$ ($N = \#$ of samples)
 2. Repeat steps 3-5 for m-th weak learner ($m = 1 \dots M$)
 3. Train the weak learner to minimize:
(error func) $E^{(m)} = \frac{\sum_{i=1}^N [\hat{y}^{(m)}(x_i) \neq y_i]}{\sum_i W_i^{(m)}}$
 4. Determine the weight for the trained learner:
 $\alpha^{(m)} = \ln \left(\frac{1 - E^{(m)}}{E^{(m)}} \right)$ (“the amount of say”)
 5. Update the weights for training data: $w_i^{(m+1)} = \dots$

Implication:

1. if $E^{(m)} < 0.5$, then $\frac{1 - E^{(m)}}{E^{(m)}} > 1$, resulting in $\alpha^{(m)} > 0$
better learners obtain higher positive voting weight
2. Given $y_i \in \{-1, 1\}$ and $\hat{y}^{(m)} \in \{-1, 1\}$
Case A: $y_i = \hat{y}^{(m)}(x_i) \rightarrow \hat{y}^{(m)}(x_i)y_i = 1$ (**Correct Case**) (Low Bias)
then $w_i^{(m+1)} = w_i^{(m)} \exp(-\alpha_m)$
since $\alpha_m > 0$, $\exp(-\alpha_m) < 1 \rightarrow$ **weight decreases**
Case B: $y_i \neq \hat{y}^{(m)}(x_i) \rightarrow \hat{y}^{(m)}(x_i)y_i = -1$ (**Highly biased**)
then $w_i^{(m+1)} = w_i^{(m)} \exp(+\alpha_m)$
since $\alpha_m > 0$, $\exp(+\alpha_m) > 1 \rightarrow$ **weight increases**

*Each learner targets the localized bias of the previous ensemble. Sequential.
Other General Boosting doesn't use weighted average.*

More About Boosting

Ensemble Methods: Foundations and Algorithms, Zhi-Hua Zhou

General procedure

Input: Sample Dist: D , Base learning algo: L , # of learning rounds T

1. $D_1 = D$
2. for $t = 1 \dots T$
3. $h_t = L(D_t)$ (train a learner)
4. $\epsilon_t = P_{x \sim D_t}(h_t(x) \neq f(x))$ (evaluate the error)
5. if $\epsilon_t > 0.5$ then break
6. $\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$ - odds ratio.
7. $D_{t+1}(x) = \frac{D_t(x)}{Z_t} \times \begin{cases} \exp(-\alpha_t) & \text{if } h_t = f \\ \exp(\alpha_t) & \text{if } h_t \neq f \end{cases}$
 $= \frac{D_t(x) \exp(-\alpha_t \cdot f(x) \cdot h_t(x))}{Z_t}$
8. end

Output: $H(x) = \text{Combine-Outputs}(\{h_1(x), \dots, h_T(x)\})$

Ada Boost: Additive model, gradient descent on an Exponential loss (only for classification) $[-1, 1]$

Output: $\text{sign}(H(x))$ with model function: $H(x) = \sum_{t=1}^T \alpha_t h_t(x)$

$H(x)$ = continuous real number

$|H(x)|$ = margin (physical distance of point to the model's decision boundary)

- Sign of margin = correct / incorrect
- Value of margin = confidence (high/low)

$$(e^u)' = e^u \cdot u'$$

Now, exponential loss $l_{exp}(H) = e^{-yH}$

then the expected exponential loss over a data distribution is:

$$l_{exp}(H|D) = E_{x \sim D}[e^{-f(x)H(x)}]$$

$$\frac{\partial l_{exp}(H|D)}{\partial H(x)} = -e^{-H(x)}P(f(x) = 1|x) + e^{H(x)}P(f(x) = -1|x)$$

if $H(x)$ can minimize the loss, it can solve by $\frac{\partial l_{exp}(H|D)}{\partial H(x)} = 0$

$$\text{i.e., } H(x) = \frac{1}{2} \ln \frac{P(f(x)=1|x)}{P(f(x)=-1|x)}$$

$$\text{then } \text{sign}(H(x)) = \begin{cases} 1, & P(f(x) = 1|x) > P(f(x) = -1|x) \\ -1, & \text{otherwise} \end{cases} \quad (\text{ignore} = \frac{1}{2})$$

$$= \text{argmax}_{y \in \{-1, 1\}} P(f(x) = y|x)$$

Finding the optimal weight α_t :

Once the classifier h_t is generated from D_t , its weight α_t is estimated by minimizing the exponential loss.

$$l_{exp}(\alpha_t h_t | D_t) = e^{-\alpha_t} \cdot P(\text{correct}) + e^{\alpha_t} \cdot P(\text{incorrect})$$

$$= e^{-\alpha_t}(1 - \epsilon_t) + e^{\alpha_t} \cdot \epsilon_t \quad (\text{weighted error rate under } D_t)$$

Set derivative to 0 and get (line 6):

→ larger α_t when ϵ_t is small.

At iteration t , with a sequence of base learners h_1 to h_{t-1} and corresponding weights, $H_{t-1}(x) = \sum_{i=1}^{t-1} \alpha_i h_i(x)$ and AdaBoost adjusts the sample distribution such that the base learner h_t can learn from mistakes of H_{t-1} .

Deriving the dist. update

Notion Required: Taylor expansion:

$$e^u = 1 + u + u^2/2! + u^3/3! + \dots \quad (\text{don't need constant to get critical point})$$

the ideal h_t which can correct all mistakes of H_{t-1} should minimize:

$$l_{exp}(H_{t-1} + h_t | D) = E_{x \sim D}[e^{-f(x)(H_{t-1}(x) + h_t(x))}]$$

$$\rightarrow h_t(x) = \text{argmin}_h l_{exp}(H_{t-1} + h_t | D)$$

$$\text{now } E_{x \sim D}[e^{-f(x)(H_{t-1}(x) + h_t(x))}]$$

$$= E_{x \sim D}[\underbrace{e^{-f(x)H_{t-1}(x)}}_{\text{constant for } h_t} \cdot \underbrace{e^{-f(x)h_t(x)}}_u]$$

$$\simeq E_{x \sim D}[e^{-f(x)H_{t-1}(x)} \cdot (1 - f(x)h_t(x) + \frac{1}{2}f^2(x)h_t^2(x))]$$

$$\text{since } f^2(x) = h_t^2(x) = 1, (-f(x)h_t(x))^2 = f^2(x)h_t^2(x) = 1$$

dropping $u^3/3!$ form and after approximates constant - dropped

$$\simeq E_{x \sim D}[e^{-f(x)H_{t-1}(x)} \cdot (1 - f(x)h_t(x) + \frac{1}{2})]$$

$$\text{then } h_t(x) = \text{argmax}_{h_t} E_{x \sim D}[e^{-f(x)H_{t-1}(x)} \cdot f(x)h_t(x)]$$

now by definition weighted expectation means:

$$E_{x \sim D}[w(x) \cdot g(x)] = \sum_x D(x) \cdot w(x) \cdot g(x)$$

$$\text{when } D_t(x) = D(x) \cdot w(x)/Z,$$

What it did: move the w out of E into dist.

$$= Z \sum_x D_t(x) \cdot g(x)$$

$$= Z \cdot E_{x \sim D_t}[g(x)]$$

$$\text{Now apply } w(x) = e^{-f(x)H_{t-1}(x)} \text{ and } g(x) = f(x)h_t(x)$$

$$\text{and } E_{x \sim D}[e^{-f(x)H_{t-1}(x)} \cdot f(x)h_t(x)] = Z \cdot E_{x \sim D_t}[f(x)h_t(x)]$$

$$\text{Where } D_t(x) = \frac{D(x)e^{-f(x)H_{t-1}(x)}}{Z} \text{ and } Z = \sum_x D(x) \cdot e^{-f(x)H_{t-1}(x)}$$

Since Z is just a positive constant $\rightarrow$ *dropped*

$$\text{and } h_t(x) = \operatorname{argmax}_{h_t} E_{x \sim D_t} [f(x)h_t(x)]$$

here,

H_{t-1}	$f(x)H_{t-1}(x)$	$e^{-f(x)H_{t-1}(x)}$	weight
Correct/Confident	large pos.	small	low
uncertain/near zero	near zero	~ 1	medium
incorrect/confident	large neg.	large	high

$$\begin{aligned} &\text{Since } f(x)h_t(x) = 1 - 2 \cdot I[f(x) \neq h_t(x)] \\ h_t(x) &= \operatorname{argmin}_{h_t} E_{x \sim D_t} [I(f(x) \neq h_t(x))] \end{aligned}$$

Next iteration prep (recursive form)

$$D_{t+1}(x) = \frac{D_t(x) \cdot e^{-f(x) \cdot \alpha_t \cdot h_t(x)}}{Z_t}$$

which expands to the relationship between D_{t+1}, D_t (residue minimization):

$$\begin{aligned} D_{t+1}(x) &= \frac{D_t(x) e^{-f(x) H_t(x)}}{E_{x \sim D} [e^{-f(x) H_t(x)}]} \\ &= \frac{D_t(x) e^{-f(x) H_{t-1}(x)} \cdot e^{-f(x) \alpha_t h_t(x)}}{E_{x \sim D} [e^{-f(x) H_t(x)}]} \\ &= \frac{D_t(x) \cdot e^{-f(x) \alpha_t h_t(x)} \cdot E_{x \sim D} [e^{-f(x) H_{t-1}(x)}]}{E_{x \sim D} [e^{-f(x) H_t(x)}]} \\ Z_t &= \frac{\sum_x D_t(x) \cdot e^{-f(x) \alpha_t h_t(x)}}{E_{x \sim D} [e^{-f(x) H_t(x)}]} \end{aligned}$$

h_t	$f(x)h_t(x)$	$e^{-f(x)\alpha_t h_t(x)}$	$\times D_t(x)$
Correct	+1	$e^{-\alpha_t}$	$e^{-\alpha_t}$ if $h_t(x) = f(x)$
Incorrect	-1	e^{α_t}	e^{α_t} if $h_t(x) \neq f(x)$

which yields (line 7)